

in carrying out the tasks that need to be done to complete a project software.

3.3 Comparison of Software Development Methods

Comparison of software development methods is done to determine the characteristics of each software development process to facilitate software developers choose the model according to their needs. There are three models compared to this paper: V-shaped model, parallel model, and iterative model.

3.4 V-Shaped Model

V-Shaped Model called by Balaji and Murugaiyan [2] "verification and validation model" is a modified version of the Waterfall model. Unlike Waterfall, this one model is not designed linearly. If in the waterfall model the process is run linearly, then in V-model the process is done branched. Software development process in V-Shaped Model depends on verification in the previous stage. In V-Shaped Model gives the relationship between each development stage and Testing stage.

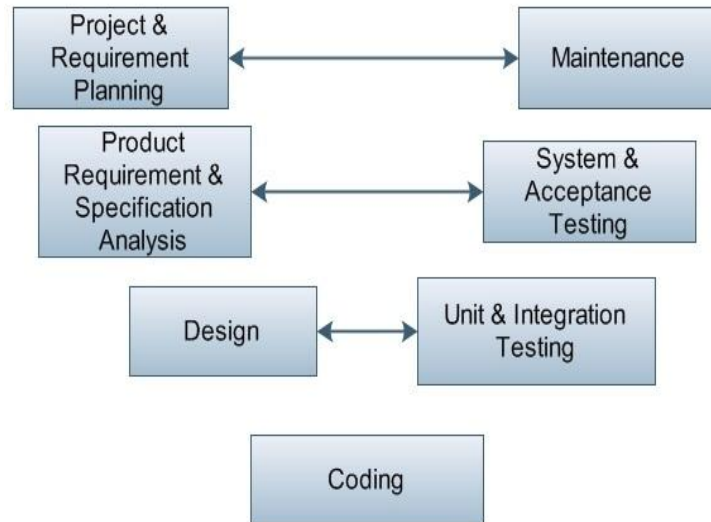


Figure 2: V-Shaped Model

Excess of V-Shaped Model

In the opinion of Munassar and Govardhan [7], V Shaped Model has advantages such as:

1. Straightforward and easy to use.
2. Each stage has a certain result.
3. The chance to succeed is higher than the waterfall model because the test is done from the beginning
4. This model is well used for software projects on a small scale

Lacks of V-Shaped Model

While the shortcomings of V-Shaped Model in the opinion of Munassar and Govardhan [7] are:

1. Very rigid as waterfall model.
2. Flexibility is minimal and when will adjust difficult and expensive

3. The software was developed during the implementation phase so that no initial prototype of the software was produced
4. This model does not provide a clear solution to the problems found during the testing phase.

3.5 Parallel Model

In the opinion of Ajah and Ugah [1], this model tries to overcome the long time interval between the period of analysis and delivery of the system. A general design for the whole system is done, and then the project is divided into a series of different sub-projects that can be developed and implemented in parallel. This model tries to overcome the weaknesses in the waterfall model so that the concept is made after all subprojects are perfect, then the final integration is done so that delivery is done to the system.

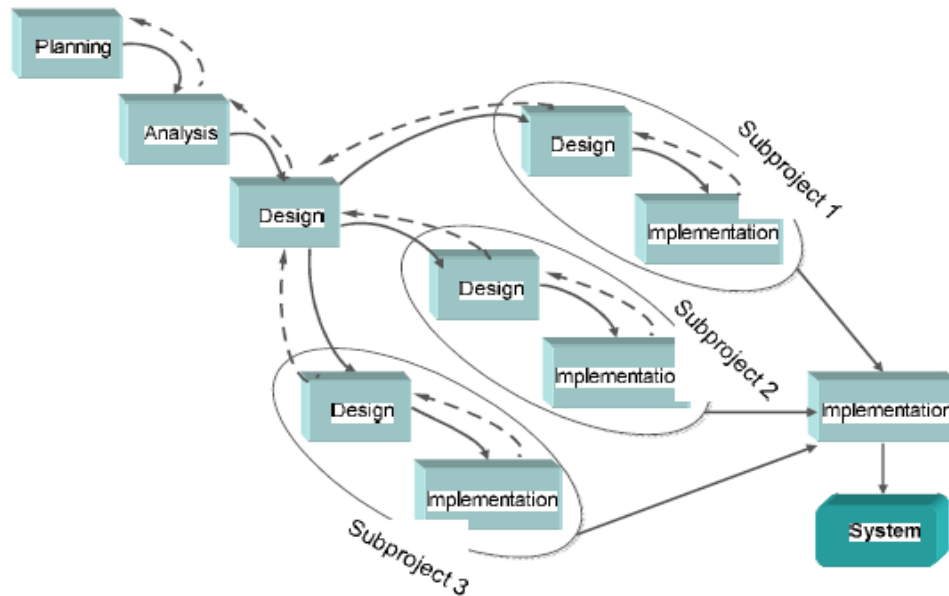


Figure 3: Parallel Model

Excess of Parallel Model

Ajah and Ugah [1] argue that the excess parallel model is the time required for system delivery is shorter when compared to the waterfall because some stages have been made in parallel form thus reducing the duration of time.

Lacks of Parallel Model

In the opinion of Ajah and Ugah [1] the parallel shortcomings of the model are:

1. Just like the waterfall design, this model also could not be separated from problems caused by long system delivery.
 2. Sometimes subprojects are not independent; this is because the design decision is made in one
 3. subproject may affect others.
 4. The end of the project may involve significant integration challenges.
 5. Not suitable for complex projects and necessary software
- While Budi, Siswa, and Abijono [4] added the shortcomings of the parallel model are:

1. System integration has its difficulties. Failure or delay in one subproject impacts the process of integrating the entire system,
2. There is a possibility of difficulty in handling if there is a problem in the subproject simultaneously.

3.6 Iterative Model

According to Munassar and Govardhan [7] problems that occur in the waterfall model create demand for software development models that can deliver faster results, require less information, and offer greater flexibility.

With an iterative design, a project is divided into small parts; This allows the development team to show the results of previous processing and gain valuable feedback from users of the system. Often, each iteration is a mini-Waterfall process with feedback from one stage and providing necessary information for the next stage.

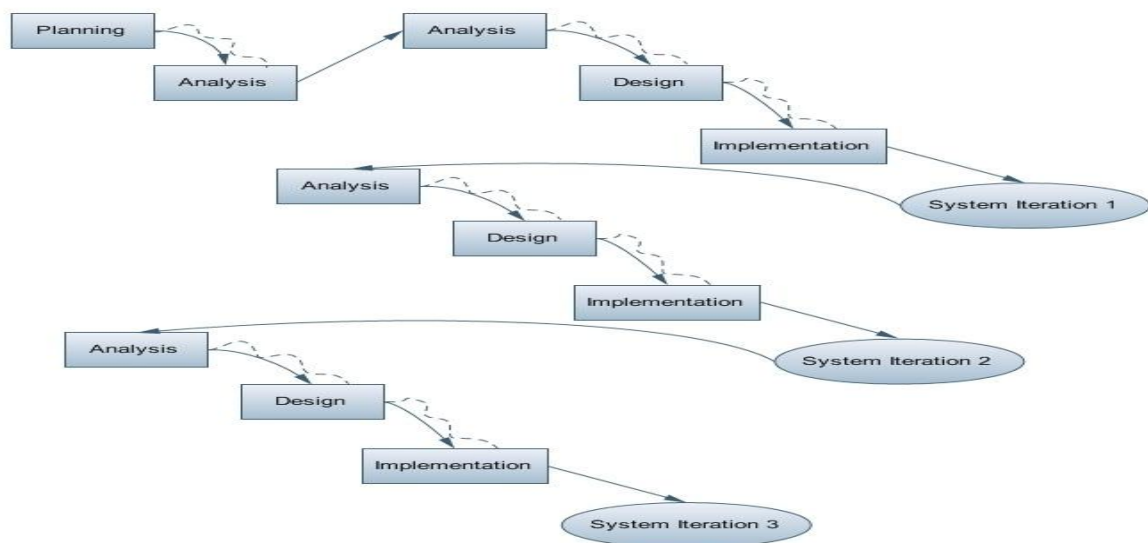


Figure 4: Iterative Model

Excess of Iterative Model

In the opinion of Budi et al. [4] the advantages of iterative models are as follows:

1. Feedback can be done continuously from the project owner.
2. Some of the revisions to all specific applications and functions.
3. Work in software development delivered at the beginning of the project

Lacks of Iterative Model

Lacks of iterative modifications include:

1. Although not necessarily a problem for all projects, since minimal initial planning before coding and implementation begins, when using the iteration model, it is possible that unexpected problems in the design or system architecture will arise within the project. Completing this could have potentially disastrous effects on the overall project time and cost of the project, which require many future iterations just to solve one problem.
2. Unlike the waterfall model, which emphasizes almost all user involvement in the early stages of the project, repetitive models often require user engagement throughout the entire process, as each new iteration will likely require testing and feedback from users to evaluate the necessary changes.

4. CONCLUSION

From comparison to 3 (three) method above, can be concluded as follows:

1. V-Shaped Model is easy to use but rigid as waterfall model, But the chance of software development success using shaped model is bigger when compared with waterfall model.
2. The time used by the parallel model in the system delivery is shorter than the waterfall model
3. The iterative design requires assistance from users for all stages of system development.
4. In v-shaped models and iterative models, the cost of system development is more expensive than the waterfall model.
5. The project owner's iterative model can perform continuous feedback.

It can be concluded that the success of software development depends on the overall management of software projects. The

software development methodology component consists of models, tools (Tools), and procedures. No model really suits all types of organizations, so it takes a further approach to choosing which model is most appropriate to apply to a particular group.

5. ACKNOWLEDGMENTS

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